

Advanced Structural Analysis using ANSYS

Lecture and Exercise Notes

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Lecture 2: Nonlinear Structural Analysis with ANSYS

Die Veranstaltung basiert u.a. auf dem Vorlesungs- und Übungsskript „Die Methode der finiten Elemente“, gehalten durch Prof. Dr. rer. nat. M. Braun, Dr.-Ing. H. Walter, Dr.-Ing. U. Zwiers und Dr.-Ing. Philipp Olle, am Lehrstuhl für Mechanik (Maschinenbau) der Universität Duisburg-Essen im WS2004/05

Recapitulation

Linear Finite Element Analysis is based on

- linearized geometrical equations: $\epsilon = Bu$
(strain-displacement relations)
- linearized constitutive equations: $\sigma = D\epsilon$, Hooke's law
(stress-strain relations)
- equations of equilibrium: $Ku = R$

and suitable boundary conditions.

Introductory Remarks on Nonlinearity

$$\text{Nonlinear effects} \Rightarrow \begin{cases} \mathbf{K} = \mathbf{K}(\mathbf{u}) \\ \mathbf{R} = \mathbf{R}(\mathbf{u}) \end{cases}$$

For nonlinear problems, it is not possible to solve $\mathbf{K}\mathbf{u} = \mathbf{R}$ immediately as $\mathbf{K}$ and $\mathbf{R}$ are not known in advance.

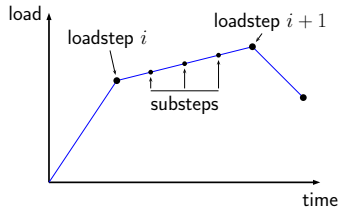
Types of Nonlinearity in Structural Mechanics:

- Geometric Nonlinearity
(large deformation)
- Material Nonlinearity
(hyper-elasticity, plasticity, creep, anisotropic behavior)
- Boundary Nonlinearity
(contact)

Introductory Remarks on Nonlinearity (cont'd)

Basic Characteristic of Nonlinear Simulations:

Results are obtained by *iterative* calculations and *increment* processing.



Nonlinear Analysis with ANSYS

Provided Iteration Schemes:

- Newton-Raphson method (default)
- Newton-Raphson method with adaptive descent
- Arc-length method

Convergence-enhancement features:

- Line search
- Automatic load stepping
- Bisection

Nonlinear Analysis with ANSYS (contd.)

ANSYS-parameter TIME:

- Time-independent analysis (statics)
TIME denotes simply a numbering parameter
- Time-dependent analysis (dynamics)
TIME denotes the real, chronological time

Performing a Nonlinear Static Analysis with ANSYS

1. Build the model

A nonlinear analysis might include special elements (e. g., contact elements) and/or nonlinear material properties.

2. Set solution controls

Allow for large deformations (NLGEOM), control data output (OUTRES), activate special solution options (PRED, LNSRCH, SSTIF)

3. Apply the loads

Specify load steps (TIME, KBC), control substepping (AUTOTS, NSUBST, DELTIM)

4. Solve the analysis

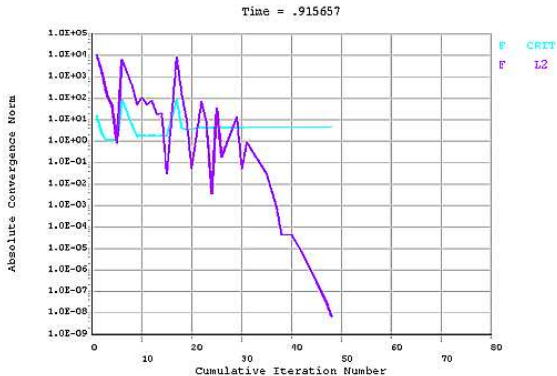
5. Review the results

/POST1: results at specific substeps (time-points)

/POST26: specific result items over the applied load history

Graphical Solution Tracking

ANSYS



Graphical Solution Tracking (cont'd)

- Title: Time value of the last calculated iteration
- Label x -axis: Cumulative Iteration Number
In nonlinear analysis, ANSYS uses a solver that iterates to find a solution. If a problem is relatively linear, very few iterations are required, while highly nonlinear problems necessitate many iterations.
- Label y -axis: Absolute Convergence Norm (non-normalized values)

The quantities being plotted are always some residual versus a tolerance (residual: difference between the norms of applied load and calculated reaction load). When the residual becomes less than tolerance, a converged solution is obtained.

ANSYS calculates a convergence criterion based on the norm of the applied load (the tolerance value can be adjusted by the CNVTOL command). The type of analysis determines the convergence quantities taken into account, e. g., force, moment, displacement, temperature,...